PrandialTwin

A digital twin for post-meal hyperglycaemia that tests its own claims

Team: TODO team name | TODO college and incubator
Happiest Health Digital Twin Challenge 2026

After-meal glucose is India's blind spot

26.7%

of Indian adults have diabetes (11.4%) or prediabetes (15.3%)

61.6%

of energy in the average Indian diet comes from carbohydrate

53.5%

of time above range in 649 Indian T2D patients on a first glucose sensor

The gap

HbA1c every three months averages the spikes away. A 14-day CGM report shows them, but only after they happened. Nobody tells the doctor or the patient before the meal, when a smaller portion or a walk could still change the outcome.

Sources: ICMR-INDIAB-17 (Lancet Diabetes Endocrinol 2023); Anjana et al. (Diabetes Care 2022); Gupta (Cureus 2025)

Most "digital twins" are not twins

1. **It updates**

   Dynamically updated with data from the real patient

2. **It predicts**

   Forecasts what happens next, with quantified uncertainty

3. **It informs a decision**

   Answers a what-if a clinician can act on, and shows its track record

National Academies (2023) definition, with verification, validation and uncertainty quantification built in

12%

of 149 published health "digital twins" meet that definition

1.3%

mention verification, validation and uncertainty
Tudor et al., npj Digital Medicine 2025

What the doctor asks, and what the twin answers

The doctor asks	The twin gives
Will this meal push her above 180 mg/dL?	Probability of glucose above 180 mg/dL for 15+ minutes in the next 2 to 3 hours, with a calibrated band
Why?	How much carbohydrate on board, current drift and recent activity add to the expected peak
What would help?	The same forecast re-run with a smaller portion, another meal, or a walk
Can I trust that?	The track record of such what-ifs on real matched meals, shown beside the answer
Is today unusual for her?	Today's carbohydrate response against her own baseline

Adverse event predicted: CGM above 180 mg/dL for at least 15 consecutive minutes, forecast while glucose is still at or below 180.

Two streams, one person, and real data first

CGMacros

45 real adults, USA

Labs + Dexcom CGM + Fitbit + meals with macros, about 10 days

Role: development, 5-fold by patient

PhysioNet, CC BY-NC-SA 4.0

ShanghaiT2DM

109 real recordings, China

15-minute CGM, meals as text, half on insulin, no wearable

Role: unseen test only

figshare, CC BY 4.0

Synthetic cohort

60 virtual patients, Indian meals

EHR tables + CGM, heart rate, steps, sleep from one hidden physiology

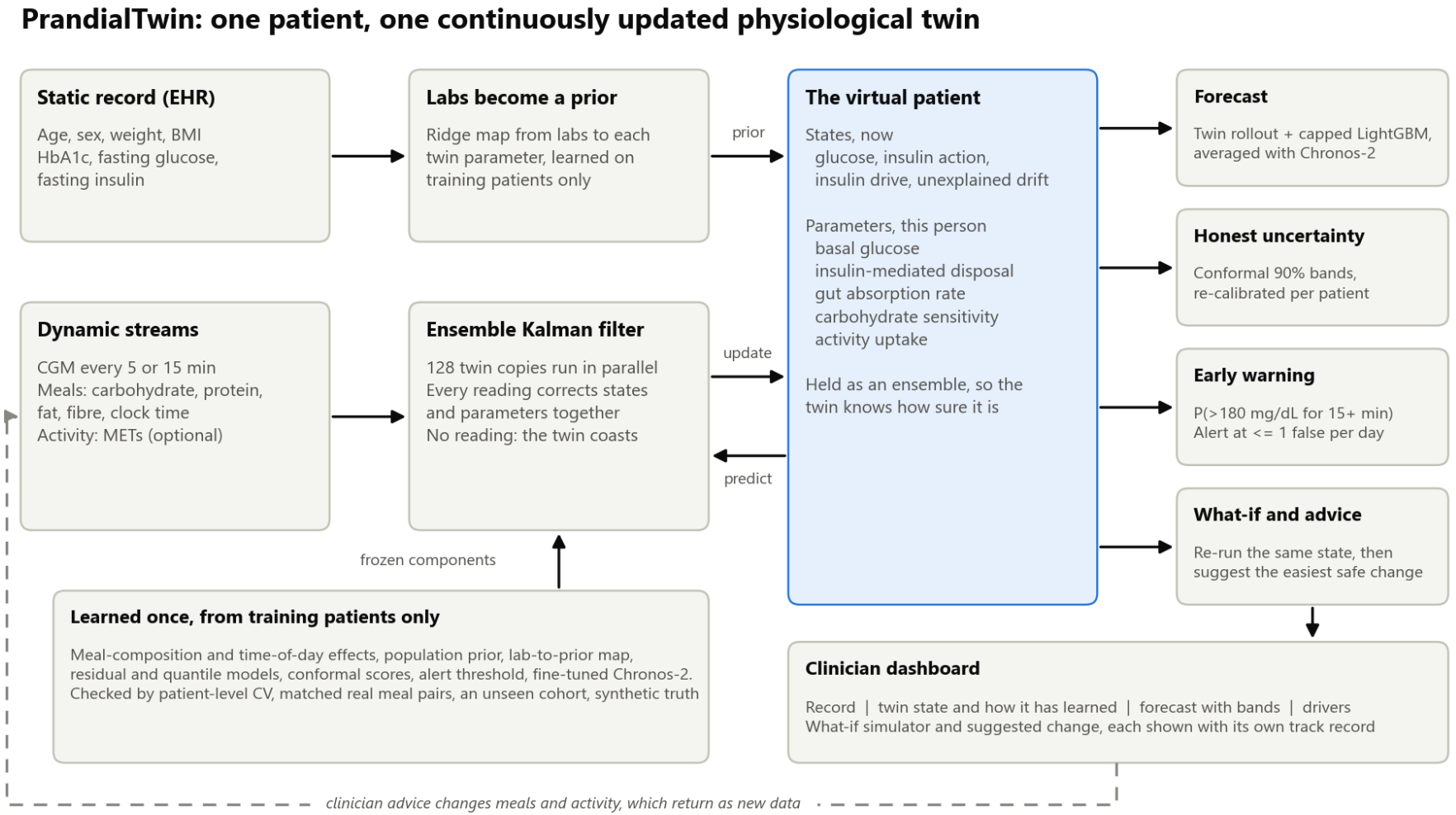
Role: verification and demo only

Generated here

Static record: age, sex, weight, BMI, HbA1c, fasting glucose, fasting insulin. Dynamic: CGM, meals, activity.

No Indian CGM dataset with linked labs is openly available. We say so, and do not claim Indian accuracy.

Architecture: labs set the prior, every reading updates the twin



A small physiology model, kept in step with the patient

The virtual patient

States, now: glucose, insulin action, insulin drive, unexplained drift

Parameters, this person: basal glucose, insulin-mediated disposal, gut absorption rate, carbohydrate sensitivity, activity uptake

Shared by everyone: meal-composition and time-of-day effects

1

Labs set the prior

HbA1c, fasting glucose, fasting insulin and BMI map to a starting estimate of each parameter

2

Every reading corrects it

A 128-member ensemble Kalman filter updates states and parameters together; it coasts through sensor gaps

3

Roll forward three hours

Physics forecast, a capped LightGBM correction, and a conformal band re-calibrated per patient

Never fitted to its own patient: prior, then online updates only

What the clinician sees

1

Record and twin state

Labs, basal glucose, standard-meal peak, today vs own baseline

2

Forecast and alert

Trajectories, 90% band, probability of the event, alert threshold

3

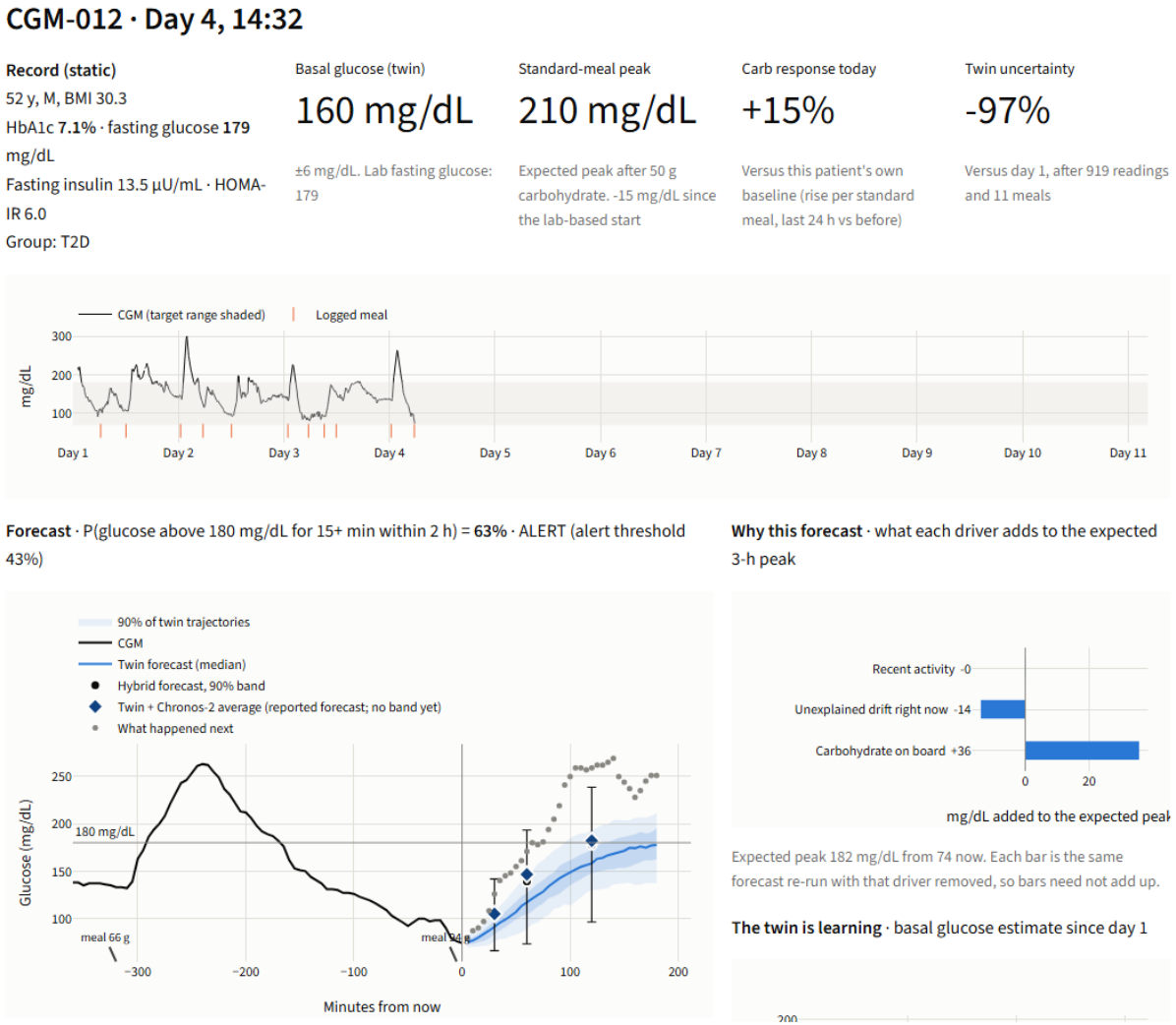
Why

Carbohydrate on board, drift, activity

4

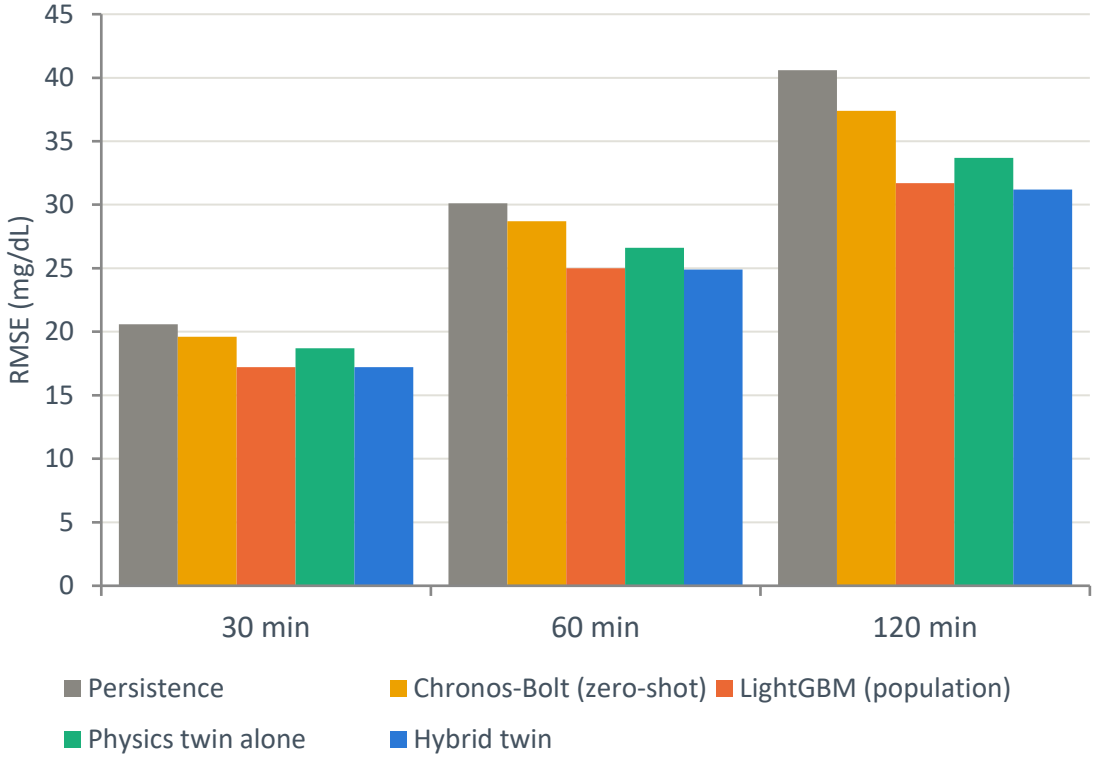
What if

Portion, extra meal, walk, with the track record shown

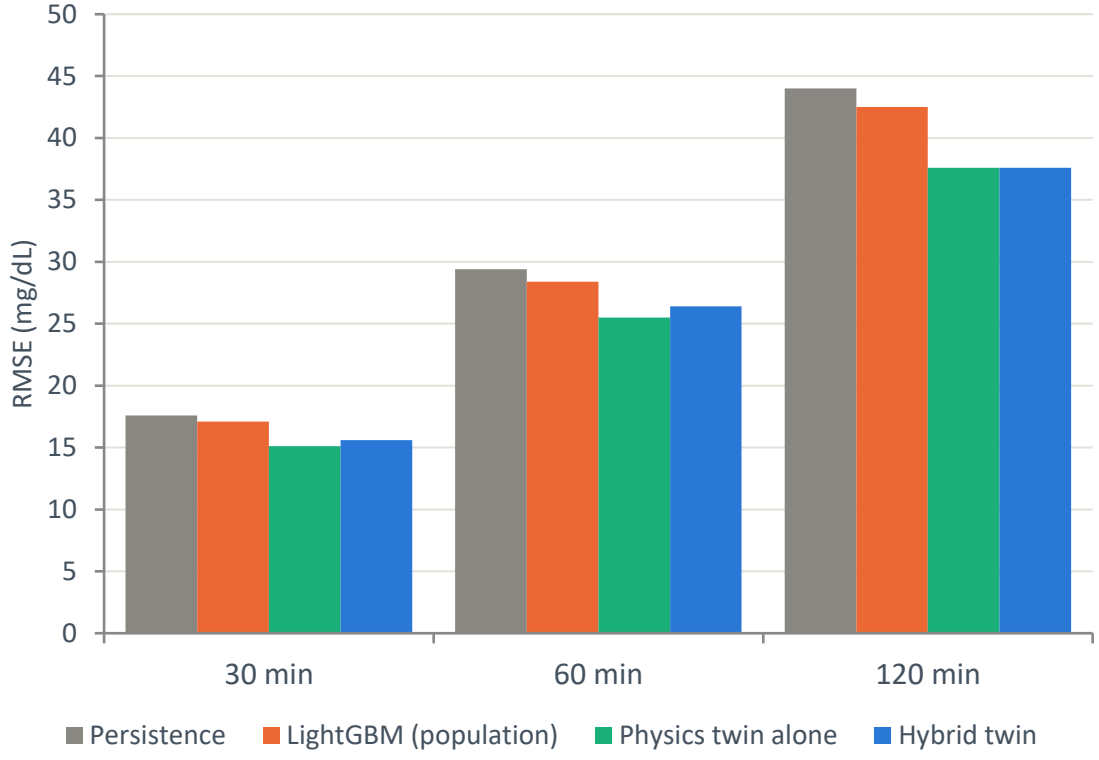


At home it ties LightGBM; away from home it beats it

Unseen patients, same cohort (CGMacros)

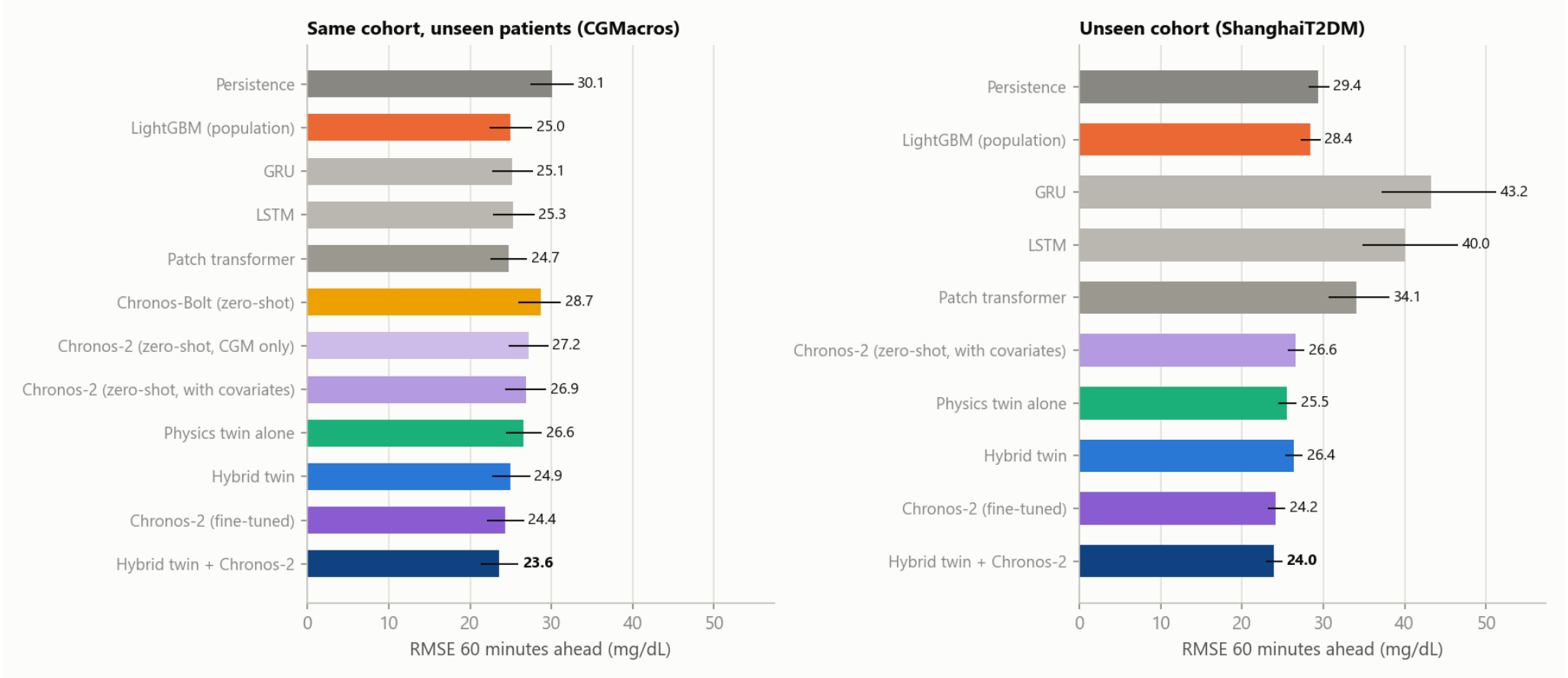


Unseen country, device and population (ShanghaiT2DM)



Hybrid minus LightGBM at 60 min: -0.02 (-0.71 to 0.57) mg/dL at home, -1.99 (-2.64 to -1.39) on the unseen cohort (95% interval, resampling patients).

We tested every model we had ruled out



60-minute RMSE, 95% intervals by resampling patients. Small networks tie at home and fail abroad. Fine-tuned Chronos-2 is the best single model in both cohorts. The twin averaged with Chronos-2 is the most accurate forecast: 23.6 at home, 24.0 abroad.

Why the networks failed abroad, and Chronos-2 did not

+19

mg/dL mean error of the GRU abroad, too high at every hour of the day. The physics twin: +1

51%

of Shanghai recordings are on insulin, a drug the networks never saw; their errors are largest there

24.2

mg/dL for fine-tuned Chronos-2 abroad, against 26.4 for the hybrid twin. It reads a day of the patient's own data

Read it carefully

Small networks learn one population's glucose behaviour, and nothing in them adapts to a new person. The twin re-learns each patient from their readings; Chronos-2 conditions on the patient's last 24 hours. Averaged, they forecast best.

It warns of more events at the same alert burden

61%

of 693 real events warned of before onset, at 0.94 false alerts per patient-day. Standard model: 52%

30 min

median warning before glucose stays above 180 mg/dL

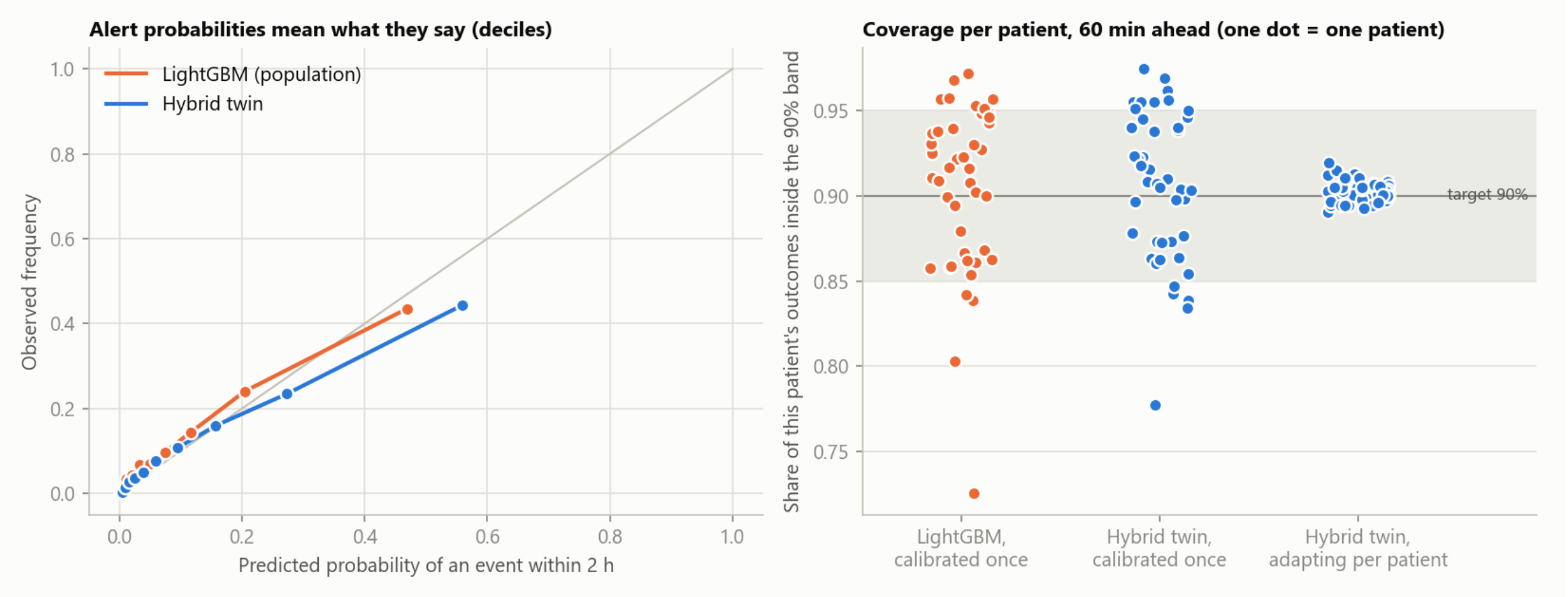
0.45

alert precision-recall area on the unseen cohort. Standard model: 0.36; difference 0.09 (0.06 to 0.14)

Read it carefully

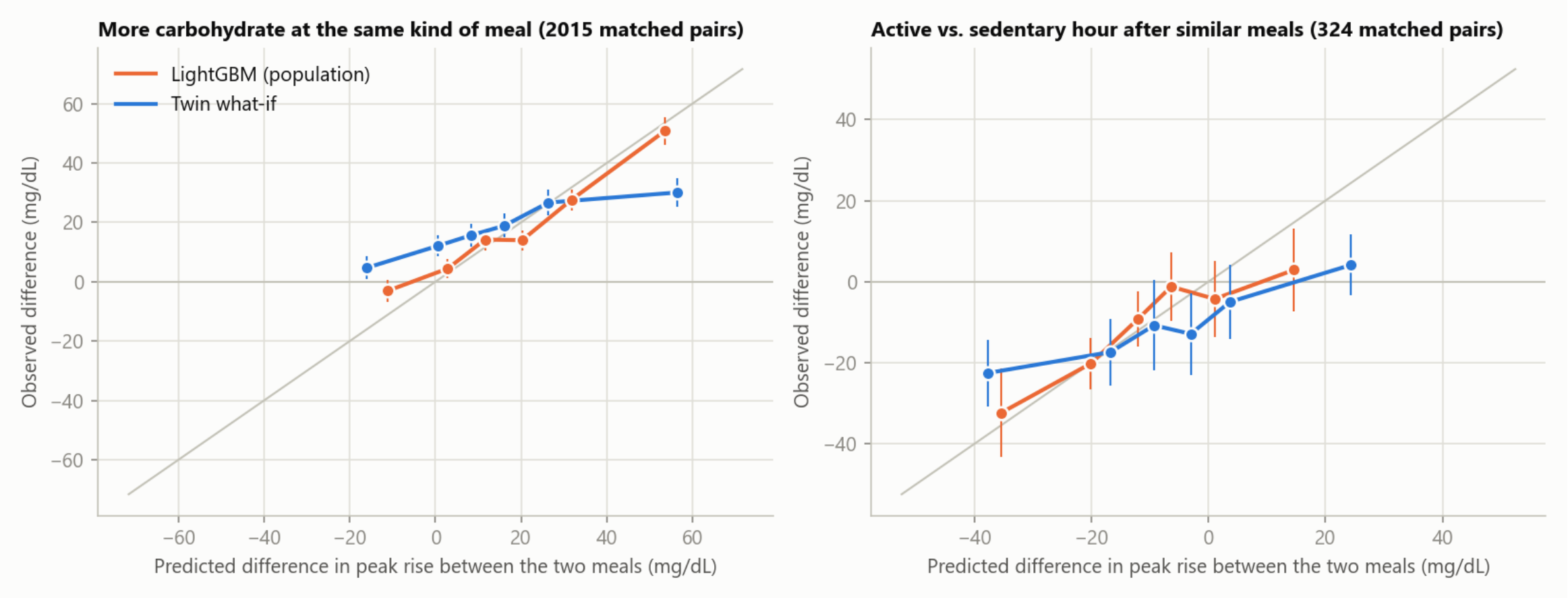
Home cohort: alert AUROC 0.82 vs 0.81, precision-recall area 0.43 vs 0.42: a tie. Only forecasts made while glucose is still at or below 180 mg/dL are counted, which is harder than detecting a spike already under way.

Its uncertainty is honest for every patient



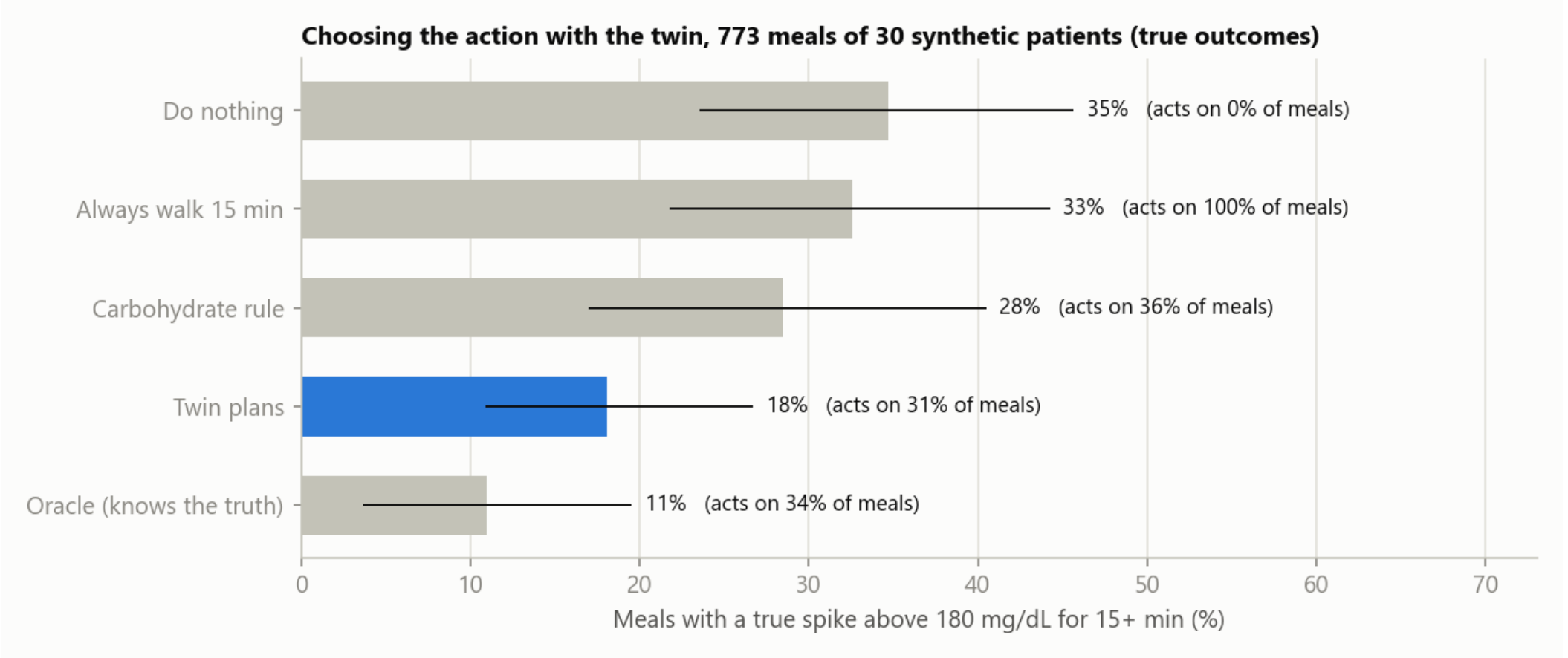
Worst patient, 60 min ahead: 89% with per-patient adaptation vs 78% without. On the unseen cohort: 88% vs 64%.

What-ifs were checked against what really happened



Direction right: 66% of 2,015 carbohydrate pairs, 61% of 324 activity pairs. Active hour after a meal: observed -10.7 mg/dL, twin -9.1. Not better than LightGBM on direction (68%, 63%).

Twin-chosen changes nearly halve spikes in simulation



773 meals, 30 synthetic patients, true outcome of every action known. Twin minus doing nothing: -16.6 (-23.0 to -10.0) points; minus a carbohydrate rule: -10.3 (-17.5 to -3.5). An upper bound, not a trial.

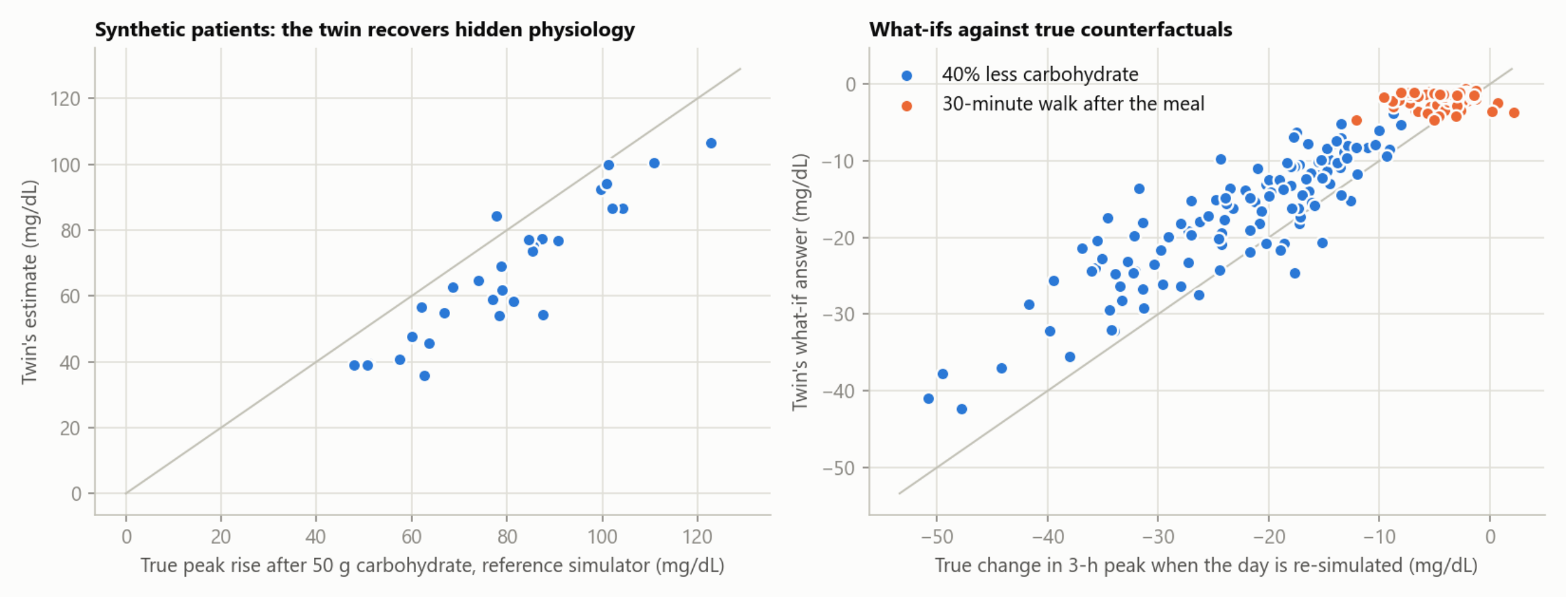
What did not hold up

We hoped for	We found
Beat LightGBM on the home cohort	A tie: 24.9 vs 25.0 mg/dL at 60 min
Labs give a clear head start	First 24 h: -0.46 (-1.01 to 0.10) mg/dL. Not significant
The twin keeps improving with data	Updating vs frozen from day 4: -0.87 (-2.14 to 0.26). Not significant; worse in the first 3 days
Twin what-ifs beat an associative model	No: 66% vs 68% direction right on carbohydrate pairs
The twin transfers better than any model	No: fine-tuned Chronos-2 24.2 vs hybrid 26.4 mg/dL abroad. Now averaged in

What did hold

Personalised vs generic physiology from day 4: -5.6 (-9.4 to -2.1) mg/dL. Twin basal glucose vs lab fasting glucose: rho 0.62. Standard-meal peak vs HbA1c: rho 0.60, with the labs withheld from the twin.

On synthetic patients, the twin recovers the truth



Basal glucose: rho 0.959, error 4.7 mg/dL. "40% less carbohydrate": r = 0.875, direction right 100%, true -22.9 vs twin -17.2 mg/dL. No accuracy claim rests on synthetic data.

Limits and responsible use

Not Indian data

45 people from the USA, 100 from China. The India case is problem fit, not measured accuracy

Retrospective

Replayed recordings. No prospective use, no clinical outcomes

No drug or insulin model

Their effect lands in the drift term. Half the external cohort is on insulin

Meals must be logged

With no meal log the 60-minute error rises by 8%

Decision support only

For a clinician. Lifestyle scenarios; no medication or dosing advice

Data terms

CGMacros is non-commercial (CC BY-NC-SA). No raw data in the repository

What comes next for India

1	**Indian validation**	Same protocol on an Indian CGM cohort with linked labs, with a hospital partner
2	**Therapy in the model**	Metformin, sulfonylurea and insulin effects; sleep and stress from the wearable
3	**Meals without typing**	Photo-based logging for Indian dishes; the twin already tolerates 30% carbohydrate error
4	**Fit the health system**	FHIR and ABDM records in and out
5	**Prospective pilot**	Alerts and what-ifs in clinic, measured against time in range

A twin you can inspect, reproduce and challenge

It beats LightGBM abroad, pairs with Chronos-2 for the best forecast, keeps honest bands, and its what-ifs and advice were checked

One command reproduces every number: python -m twin.evaluate
Repository: TODO public GitHub link